# Exploring 3D Cardiac Anatomy with Text-Based AI Guidance in Virtual Reality

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***Abstract*—This paper presents the design and development of a social virtual reality (VR) classroom focused on cardiac anatomy education for students in grades K-12. The application allows multiple learners to explore a detailed 3D heart model within an immersive and collaborative environment. A crucial part of the system is the integration of a text-based conversational AI interface powered by ChatGPT, which provides immediate, interactive explanations and addresses student inquiries about heart anatomy. The system supports both guided and exploratory learning modes, encourages peer collaboration, and offers personalized support through natural language dialogue. We evaluated the system's effectiveness through a comprehensive study measuring learning perception (LPQ), VR perception (VRPQ), AI perception (AIPQ), and VR-related symptoms (VRSQ). Potential applications include making high-quality cardiac anatomy education more affordable for K-12 schools with limited resources, offering an adaptable AI-based tutoring system for students to learn at their own pace, and equipping educators with an easy-to-use tool to integrate into their science curriculum with minimal additional training.**

***Index Terms*—social virtual reality, collaborative learning, generative AI, anatomy education**

## I. Introduction

In light of the popularity of ChatGPT and other artificial intelligence (AI) tools, their use has become increasingly common in daily life and education. Integrating these technologies into immersive learning environments offers new opportunities to support and engage both teachers and students. This study explores the integration of generative AI and virtual reality (VR) by embedding an AI tutor within a virtual classroom. Indeed, research has shown significant improvements in task performance within these immersive environments compared to traditional learning methods [1]. The potential of immersive technologies in education has also been explored by comparing virtual classrooms with e-learning platforms, further highlighting their versatility [2]. Moreover, collaborative VR environments can foster knowledge retention and create a more dynamic learning atmosphere [3]. Simultaneously, AI, particularly through conversational agents, can provide personalized learning support, offering explanations and guidance tailored to individual student needs [4]. The convergence of these technologies has the potential to create highly engaging, interactive, and effective learning environments for students in K-12 education.

Despite the recognized potential of VR and AI in education, there remains a need for learning environments that effectively integrate these technologies to support collaborative and intelligently guided exploration of complex subjects, such as cardiac anatomy, for K-12 students. Similarly, researchers have developed AI tutors, but they have rarely integrated them into immersive, multi-user VR environments tailored to specific K–12 science curricula. A significant research gap exists in the design, development, and evaluation of systems that combine social VR—enabling multiple students to learn together in a shared virtual space—with sophisticated, real-time conversational AI guidance specifically tailored for K-12 students tackling the complexities of cardiac anatomy. There is a particular need to understand how K-12 students perceive and interact with such integrated systems and their impact on learning-related outcomes.

This paper introduces a system designed to address the challenges mentioned earlier. Our solution comprises a social VR classroom where multiple K-12 students can collaboratively explore a detailed, interactive 3D model of the human heart. The environment supports both guided learning pathways and opportunities for self-directed exploration, encouraging peer interaction and discussion. A key element of the system is an AI tutor: a text-based conversational assistant powered by ChatGPT. This AI tutor offers real-time explanations of the heart's structures and functions, responds to questions asked by students, and provides supportive context, ultimately tailoring the learning experience within a collaborative environment.



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The main contributions of this paper are as follows:

1) The design and development of a social VR learning environment specifically tailored for K–12 students to explore and learn heart anatomy, featuring the integration of a text-based conversational AI (ChatGPT) for real-time, interactive guidance.
2) A comprehensive evaluation of the system with K–12 students, using a multi-dimensional framework that includes the Learning Perception Questionnaire (LPQ), VR Perception Questionnaire (VRPQ), AI Perception Questionnaire (AIPQ), and VR Symptoms Questionnaire (VRSQ) to assess its impact on student perceptions and overall usability.
3) The provision of initial insights into the feasibility, student reception, and educational potential of combining social VR with widely available large language model–based AI to teach complex anatomical topics to a K–12 audience, offering guidance for future research and development.

The remainder of this paper is structured as follows: Section 2 reviews related work in virtual reality (VR) for K-12 anatomy education, social VR for collaborative learning, and artificial intelligence (AI) in K-12 educational contexts. Section 3 outlines the system architecture, design principles, and implementation of our social VR classroom, as well as the integrated AI tutor. Section 4 describes the methodology employed for our user study with K-12 students, including participant recruitment, experimental procedure, and data collection instruments. Section 5 presents the results of this evaluation, focusing on student perceptions and usability. Section 6 discusses the implications of our findings, addresses limitations, and suggests directions for future research. Finally, Section 7 concludes the paper.

## II. Related Work

This section reviews relevant literature in three key areas: the use of VR in K–12 anatomy education, the application of social VR for collaborative learning, and the integration of artificial intelligence (AI), particularly conversational agents, in K–12 educational contexts. We summarize existing findings, highlight current limitations, and position our work within this broader research landscape.

### A. VR in K–12 Anatomy Education

VR can provide K-12 students with interactive, hands-on experiences that make abstract anatomical concepts more concrete and engaging. Studies [1], [5] have shown that immersive VR environments enable learners to visualize complex 3D structures, such as the human skeletal system or organs, from multiple perspectives, leading to improved comprehension compared to traditional 2D resources, including textbooks or videos [6], [7]. However, the deployment of VR in K-12 settings presents several challenges. Historically, the cost and accessibility of VR hardware have been significant barriers, although these issues are gradually diminishing. [8]. In particular, for younger users, the risk of cybersickness (VR-related symptoms) is a significant concern. To address this, VR experiences need to be carefully designed to minimize discomfort. [9], [10]. Our work addresses these gaps by developing age-appropriate content for cardiac anatomy within a VR environment specifically designed for K-12 users and by systematically evaluating VR-related symptoms through the VRSQ [11].

### B. Social VR and Collaborative Learning in Educational Contexts

Social VR environments, which allow multiple users to interact in a shared virtual space, offer promising avenues for collaborative learning. Research suggests that collaborative learning in VR can enhance student motivation, promote peer-to-peer teaching, and improve communication skills. In educational contexts, social VR enables students to jointly explore virtual artifacts, solve problems together, and engage in discussions, mirroring and sometimes augmenting face-to-face collaborative activities [2]. For K-12 students, the social dimension can be particularly engaging, transforming learning into a shared adventure and fostering a sense of presence and community [12]. Despite these benefits, designing compelling collaborative learning experiences in social VR for younger students requires careful consideration and planning. Challenges include structuring tasks to ensure equitable participation, managing group dynamics in a virtual setting, and providing appropriate scaffolding for collaboration, especially when dealing with complex topics [13]. While some studies have explored social VR for general K-12 education or specific subjects, such as history or language learning [14], its application to complex science topics, like anatomy, combined with intelligent guidance, is less common. Our system contributes by creating a social VR environment where K-12 students can collaboratively explore cardiac anatomy, supported by both peer interaction and AI guidance, aiming to leverage the motivational benefits of social learning while addressing the need for structured support.

### C. Text-Based Conversational AI in Educational VR

Anatomy education in VR environments presents unique challenges that require adequate instructional support and guidance mechanisms. Recent research has extensively investigated the psychological and educational implications of virtual assistants in immersive learning contexts [15], [16]. Kim et al. [17] demonstrated that embodied virtual assistants with gestural and locomotion capabilities significantly increased user trust and engagement compared to static interfaces. Similarly, Haesler et al. [18] found that users showed a strong preference for embodied avatar interactions over voice-only interfaces when performing educational tasks, indicating the importance of visual representation in virtual learning environments. Studies have also highlighted the cognitive benefits of conversational AI assistance. Kim et al. [19] reported that voice-based AI assistants effectively reduced cognitive



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load during learning tasks, though users expressed uncertainty about interactions outside their immediate visual field. The integration of conversational AI with visual representation has been shown to enhance user confidence and foster greater trust in the learning process [20], suggesting that multimodal AI interaction approaches are particularly effective for complex educational content.

The existing literature highlights the individual potentials of VR for anatomy visualization, social VR for collaborative engagement, and AI for personalized educational support in K-12 settings. However, there is a clear opportunity to investigate the synergistic effects of integrating these three technological pillars. Our work is positioned at this intersection, proposing a novel system that combines (i) immersive 3D visualization of cardiac anatomy through VR, tailored for K-12 learners; (ii) a social VR framework that fosters collaborative exploration and peer learning; and (iii) an integrated, text-based conversational AI (powered by ChatGPT) that provides real-time, interactive guidance and answers student questions. By developing and evaluating such a system for K-12 cardiac anatomy education, this research aims to contribute new insights into how these technologies can be effectively combined to create engaging and supportive learning experiences for complex scientific topics, addressing the limitations of isolated approaches and exploring a more holistic, technologically enhanced pedagogical model. The multi-dimensional evaluation framework (LPQ, VRPQ, AIPQ, VRSQ) further allows for a nuanced understanding of student reception to this integrated approach.

## III. System Design and Implementation

### A. Architecture Overview

The architecture of the AI-enhanced VR cardiac anatomy learning system is designed to support real-time, immersive, and interactive educational experiences. It integrates components from social virtual reality, generative AI, and user interface design to create a cohesive learning platform. The system architecture is organized into three primary layers (see Figure 1):

1) **User Interface Layer:**
   - Manages user roles and access by differentiating between students and teachers.
   - Guides users through a structured sequence of scenes:
     - *Login Scene* for authentication,
     - *Home Scene* for avatar selection and user setup,
     - Entry into the *Social VR Classroom*.
   - Ensures role-based navigation throughout the application.
2) **Virtual Reality Environment Layer:**
   - Implements a shared 3D virtual space using Unity and Photon.
   - Includes a multi-user classroom scene enabling real-time collaboration.
   - Features interactive educational content, such as a 3D heart model that students can explore and manipulate.
   - Supports social interaction and engagement via avatars that enable users to see and communicate with each other.
3) **AI Assistant Layer:** We designed the AI Assistant to support intelligent and personalized interaction within the virtual classroom environment. This component utilizes a text-based user interface to facilitate natural language queries, enabling learners to engage with the system through typed input rather than voice commands. The system integrates the input module with a large language model (ChatGPT), which interprets user queries and generates contextually relevant, level-appropriate instructional responses. By enabling sustained conversational interaction, this layer promotes a deeper conceptual understanding and provides adaptive learning support tailored to individual learner needs. This approach aligns with established Intelligent Tutoring System (ITS) frameworks that demonstrate improved learning outcomes through personalized, conversational guidance [21].

All layers are interconnected through a bidirectional communication architecture, ensuring seamless data flow and synchronization among users, scenes, and the AI tutor. The system's modular structure supports scalability and future enhancements, such as integrating new learning modules or AI capabilities.

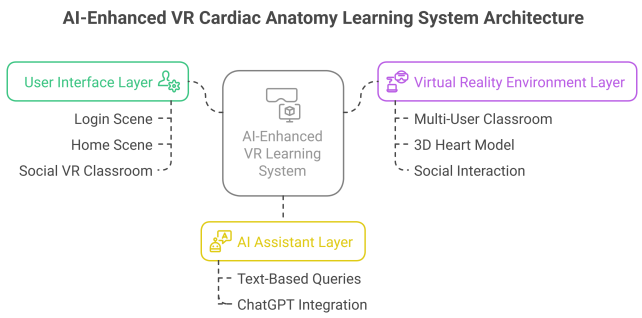


Fig. 1. System Architecture of the AI-Enhanced VR Learning Environment.

### B. AI Integration

The AI Tutor is the key to providing adaptive and tailored support in the virtual classroom. The AI Tutor, incorporated as a conversational agent within the VR platform, uses OpenAI's generative AI (ChatGPT) to understand student requests and deliver real-time educational guidance based upon each learner's needs and level of mastery.

The integration is based on two fundamental technologies:



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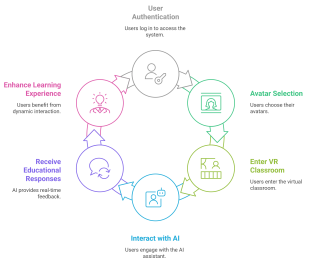


Fig. 2. User Workflow in the AI-Enhanced VR Cardiac Anatomy Learning System

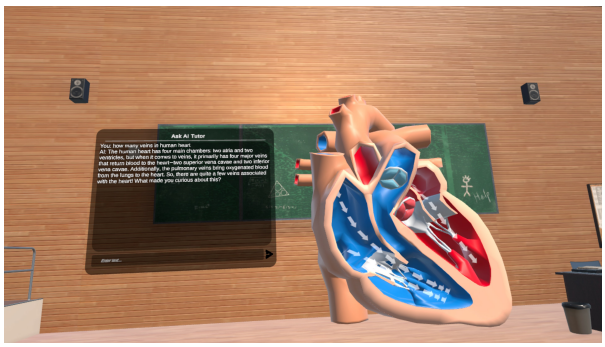


Fig. 3. AI Assisstant

1) ChatGPT through the OpenAI API, including human-like understanding and response generation.
2) A text input interface where users can type their questions and get back pedagogically sound and coherent answers.

Unlike voice communication, the system uses the text-to-text communication model, maximizing clarity and accessibility and providing easy logging of the conversational content. The students use the system directly by entering the questions in text form. They are submitted to the ChatGPT model via the OpenAI API and responded to with accurate and context-specific explanations and feedback from the AI.

The AI Tutor is embedded within the VR classroom as a dialogue interface, allowing learners to engage in natural language conversations directly within the immersive environment (Figure 3). This interface enables learners to ask questions and receive immediate, contextually relevant feedback, promoting deeper conceptual understanding and encouraging learner independence.

Key aspects of the AI Tutor integration are:

1) Real-time feedback: Tailored responses to unique users in real-time.
2) Subject Contextual Awareness: Targeted delivery of knowledge (e.g., cardiac anatomy) to support corresponding educational needs.
3) Natural Language Interaction: Engaging human-like communication stimulating further inquiry and curiosity.
4) Scalable Design: The system can simultaneously support multiple students, each receiving personalized responses.

With the OpenAI API embedded directly within the VR system, the AI Tutor can transform the classroom into a dynamic learning community where students can receive individualized instruction, enhance their knowledge, and engage in rich and meaningful dialogue.

## IV. Methodology

### A. Participants

Fifteen K-12 students participated in the evaluation study (N = 15). The sample comprised 7 males (46.7%) and 8 females (53.3%) with ages ranging from 11 to 24 years (M = 16.47, SD = 3.96). Educational backgrounds were diverse: Bachelor's degree (33.3%), high school (26.7%), middle school (20.0%), and primary school (20.0%) (see Table I for details).

TABLE I
Sample Characteristics

| Characteristic | n (%) | Age M (SD) |
|---|---|---|
| **Total Sample** | 15 (100.0) | 16.5 (4.0) |
| **Education Level** | | |
| Primary School | 3 (20.0) | 11.7 (0.6) |
| Middle School | 3 (20.0) | 13.7 (0.6) |
| High School | 4 (26.7) | 16.5 (1.3) |
| Bachelor's Degree | 5 (33.3) | 21.4 (1.7) |
| **Gender** | | |
| Female | 8 (53.3) | 16.9 (4.1) |
| Male | 7 (46.7) | 16.0 (3.9) |

### B. Procedure

Each participant took part in a single VR session, lasting approximately 30–40 minutes, conducted in a controlled classroom environment using Meta Quest 2 headsets. After a brief onboarding phase to introduce the VR controls and system features, students entered the virtual classroom environment.

Inside the VR experience, students were able to explore a detailed 3D heart model and interact with a text-based conversational AI tutor, powered by ChatGPT. The AI responded to natural language questions, providing real-time explanations related to heart anatomy. The system supported both guided and exploratory modes of learning, with opportunities for students to collaborate and ask follow-up questions (see Figure 2).

### C. Measures

To assess students' experiences and perceptions, we administered a structured post-test questionnaire immediately



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following the VR session. The instrument included four perception-focused dimensions:

1- Learning Perception Questionnaire (LPQ): Measured student interest, ease of understanding, and perceived effectiveness of technology-assisted learning.

2- VR Perception Questionnaire (VRPQ): Assessed perceptions of VR's usefulness, effectiveness compared to traditional methods, and its future adoption in education.

3- AI Perception Questionnaire (AIPQ): Evaluated clarity of AI explanations, adaptiveness to user questions, and overall engagement.

4- VR Sickness Questionnaire (VRSQ): Captured self-reported experiences of dizziness, eye strain, and disorientation during the session.

Each item was rated on a 5-point Likert scale (1 = Strongly Disagree to 5 = Strongly Agree).

## V. Results

### A. Learning and Technology Perception

Post-intervention questionnaire results demonstrated significantly positive perceptions across all measured constructs. Learning Perception (LPQ) scores were significantly above the neutral midpoint ($M = 3.96$, $SD = 0.28$), $t(14) = 13.29$, $p < .001$, $d = 3.43$, indicating strong perceived effectiveness of the cardiac anatomy learning experience. All participants (100%) agreed that interactive tools were effective for learning heart anatomy.

VR Perception (VRPQ) showed exceptionally high ratings ($M = 4.53$, $SD = 0.47$), $t(14) = 12.66$, $p < .001$, $d = 3.27$. Notably, 100% of participants rated VR integration as useful for learning cardiac anatomy ($M = 4.73$, $SD = 0.46$), and 86.7% agreed that VR was more effective than traditional 2D materials for understanding heart anatomy.

AI Perception (AIPQ) similarly demonstrated strong positive responses ($M = 4.51$, $SD = 0.40$), $t(14) = 14.76$, $p < .001$, $d = 3.81$. All participants (100%) found AI explanations clear and understandable, and rated chat-based VR sessions as highly engaging and interactive ($M = 4.80$, $SD = 0.41$).

### B. VR-Related Symptoms

VR Symptoms (VRSQ) scores were significantly below the neutral point ($M = 2.00$, $SD = 1.03$), $t(14) = -3.75$, $p = .002$, $d = -0.97$, indicating minimal adverse effects. Specifically, 86.7% of participants experienced minimal dizziness ($\leq 2$ on a 5-point scale), 73.3% reported minimal eye strain, and 66.7% experienced minimal disorientation during VR sessions.

### C. Internal Consistency

All questionnaire scales demonstrated acceptable to excellent internal consistency: LPQ ($\alpha = .71$), VRPQ ($\alpha = .78$), AIPQ ($\alpha = .75$), and VRSQ ($\alpha = .85$), supporting the reliability of the measurement instruments.

## VI. Discussion

### A. System Effectiveness and User Acceptance

The results provide compelling preliminary evidence for the effectiveness of integrating conversational AI with VR for cardiac anatomy education. The unanimous positive ratings for VR usefulness and AI clarity suggest that the combination of immersive 3D visualization with intelligent tutoring creates a highly engaging learning environment. The large effect sizes (Cohen's $d > 3.0$) across all perception measures indicate not merely statistical significance, but substantial practical impact on user experience.

The finding that 100% of participants rated interactive tools as effective for learning aligns with constructivist learning theories, which emphasize the importance of active engagement in knowledge construction. The VR environment's ability to provide spatial understanding of complex cardiac structures, combined with AI's capacity for personalized explanations, appears to address multiple learning modalities simultaneously.

### B. Minimal Cybersickness and Safety Considerations

Our results demonstrate that modern VR technology, when properly implemented, can achieve minimal adverse effects in educational contexts. The significantly low symptom scores (M = 2.00) and the fact that 86.7% of participants experienced minimal dizziness suggests that the system is suitable for extended educational use across diverse age groups. The low cybersickness rates may be attributed to several design considerations: (1) the stationary nature of the cardiac anatomy exploration, (2) the integration of text-based AI interaction that provides natural break points, and (3) the collaborative classroom environment that may reduce isolation-induced discomfort.

### C. AI Integration and Pedagogical Implications

The unanimous positive response to AI explanation clarity and engagement highlights the potential of large language models in educational VR contexts. Unlike traditional educational software with pre-scripted responses, the ChatGPT integration enables dynamic, contextual explanations that can adapt to individual student questions about cardiac anatomy. This flexibility appears particularly valuable for accommodating the diverse educational backgrounds represented in our K-12 sample. The high engagement ratings (M = 4.80) for AI interactions suggest that conversational interfaces may help bridge the gap between immersive visualization and conceptual understanding. Students can explore the 3D heart model visually while simultaneously receiving verbal explanations, creating multimodal learning experiences that support different learning preferences.

### D. Limitations and Future Directions

Several limitations need to be noted. The small sample size (N = 15) constrains generalizability, and the lack of a control group does not allow for direct comparison with



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conventional instruction. The study also measured perceptions rather than learning gains that should be remedied with future studies through pre-post tests of knowledge and longer-lasting retention studies. Future studies should incorporate randomized controlled trials comparing the VR-AI system to conventional anatomy instruction, optimal session times to balance learning gains against fatigability, and research on the adaptive AI tutoring strategies as a function of learning progression. The incorporation of biometric feedback would also yield objective measures of engagement and cognitive loading. E. Implications for Educational Technology The results add to the evidence base for the use of immersive technologies in the STEM education space. The union of VR and AI is an exciting new paradigm that combines the spatial affordances of the virtual world with the adaptability of artificial intelligence. For K-12 education in particular, with engagement and accessibility as the principal concerns, such a system can provide a means of access to more inclusive and effective science education. The universal positive reception across different educational backgrounds bodes well for the extension of such an approach to a diversity of educational contexts and domains other than cardiac anatomy.

## VII. CONCLUSION

A preliminary study has demonstrated that social VR classrooms enhanced with conversational artificial intelligence produce highly engaging and well-tolerated learning environments for cardiac anatomy. In addition to providing spatial understanding and personalized conceptual support, immersive 3D visualization combined with intelligent tutoring appears to address key challenges in K-12 science education. These results suggest promising directions for the integration of emerging technologies into educational practices, although further research with larger samples and learning outcome measures is required.

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